Part 1: Data Wrangling

```
#Load Dataset and appropriate libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

df=pd.read_csv('/content/StudentPerformanceFactors.csv')
```

```
# Inspect data
df.head()
df.info()
df.isnull().sum()

#drop NA
df.dropna(inplace=True)

df.isnull().sum()
```

```
# rename columns for further information
df = df.rename(columns={'Hours_Studied':'Hours_Studied_Per_Week',
                        'Attendance': 'Attendance_Percentage',
                        'Sleep_Hours':'Avg_Hrs_Sleep_Per_day',
                        'Physical_Activity':'Number_of_Exercise_Sessions_Per_Week',
                        'Distance_From_Home':'Distance_From_School'
                        })
df.head()
df.info()
```

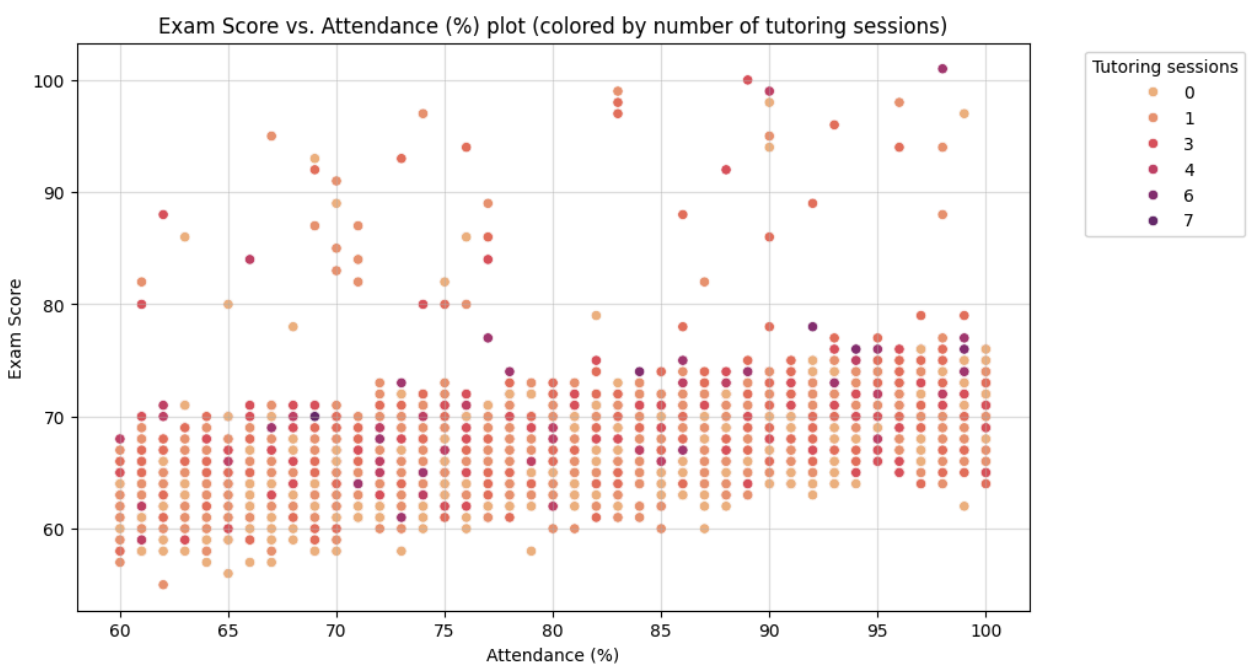
```
<class 'pandas.core.frame.DataFrame'>
Index: 6378 entries, 0 to 6606
Data columns (total 20 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Hours_Studied_Per_Week                6378 non-null   int64
 1   Attendance_Percentage                 6378 non-null   int64
 2   Parental_Involvement                  6378 non-null   object
 3   Access_to_Resources                   6378 non-null   object
 4   Extracurricular_Activities            6378 non-null   object
 5   Avg_Hrs_Sleep_Per_day                 6378 non-null   int64
 6   Previous_Scores                       6378 non-null   int64
 7   Motivation_Level                      6378 non-null   object
 8   Internet_Access                       6378 non-null   object
 9   Tutoring_Sessions                     6378 non-null   int64
 10  Family_Income                         6378 non-null   object
 11  Teacher_Quality                       6378 non-null   object
 12  School_Type                           6378 non-null   object
 13  Peer_Influence                        6378 non-null   object
 14  Number_of_Exercise_Sessions_Per_Week  6378 non-null   int64
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6607 entries, 0 to 6606
Data columns (total 20 columns):
 #   Column                      Non-Null Count  Dtype
___ ___________________________ __________________ _____
 0   Hours_Studied               6607 non-null   int64
 1   Attendance                  6607 non-null   int64
 2   Parental_Involvement        6607 non-null   object
 6   Previous_Scores             6607 non-null   int64
 7   Motivation_Level            6607 non-null   object
 8   Internet_Access             6607 non-null   object
 9   Tutoring_Sessions           6607 non-null   int64
 10  Family_Income               6607 non-null   object
 15  Learning_Disabilities       6378 non-null   object
 16  Parental_Education_Level    6378 non-null   object
 17  Distance_from_Home          6378 non-null   object
 18  Gender                      6378 non-null   object
 19  Exam_Score                  6378 non-null   int64
dtypes: int64(7), object(13)
memory usage: 1.0+ MB
```

```
# save new dataset to csv
df.to_csv('cleaned_data.csv', index=False)
```

Part 2: Visualization Creation

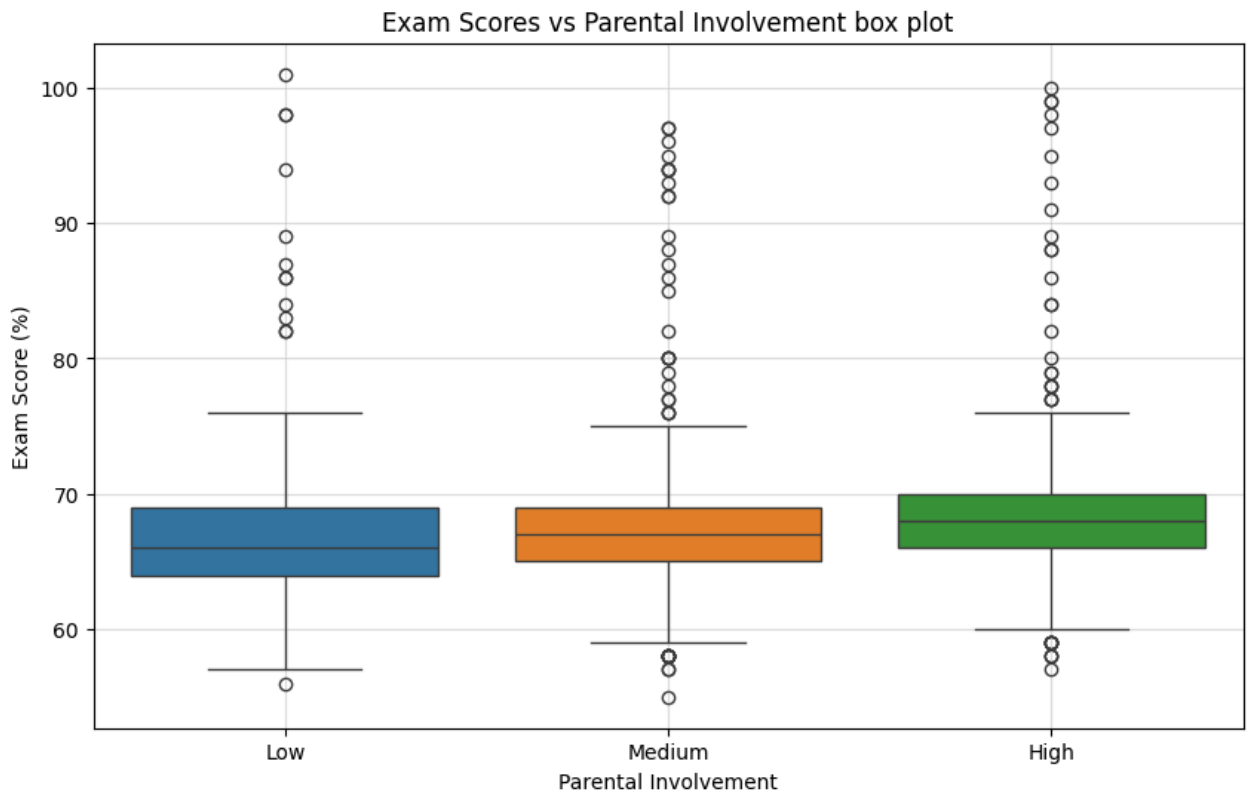
```
# Visualization 1: Scatter Plot- Exam Score vs. Hours Studied (colored by average atte
'''
Code explanation:
This code uses seaborn to create a scatter plot of exam scores vs. hours studied per w
the relationship between the number of tutoring sessions and exam scores.
'''
plt.figure(figsize=(10, 6))
scatter= sns.scatterplot(data=df, x='Attendance_Percentage', y='Exam_Score', hue='Tuto
scatter.set_title('Exam Score vs. Attendance (%) plot (colored by number of tutoring s
scatter.set_xlabel('Attendance (%)')
scatter.set_ylabel('Exam Score')
scatter.legend(title='Tutoring sessions', bbox_to_anchor=(1.05, 1), loc='upper left')
scatter.grid(alpha=0.4)
plt.show()
```



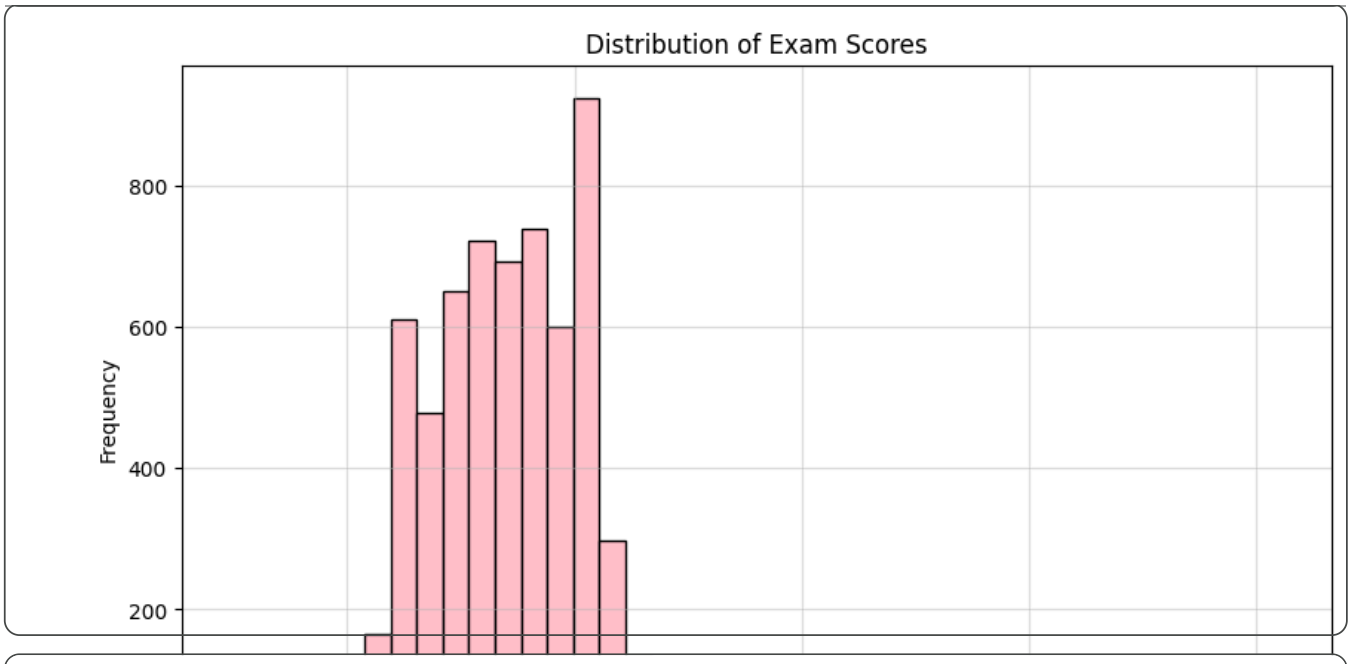

```
Learning_Disabilities      0
Parental_Education_Level   0
```

```
# Visualization 2: Box Plot- Exam Scores vs Parental Involvement
'''
Code explanation:
This code uses seaborn to create a box plot of exam scores vs. parental involvement t
```

```
exam scores given different levels of parental involvement. Each level of parental in
'''
plt.figure(figsize=(10, 6))
box= sns.boxplot(data=df, x='Parental_Involvement', y='Exam_Score',palette='tab10',hu
box.set_title('Exam Scores vs Parental Involvement box plot')
box.set_xlabel('Parental Involvement')
box.set_ylabel('Exam Score (%)')
box.grid(alpha=0.4)
plt.show()
```

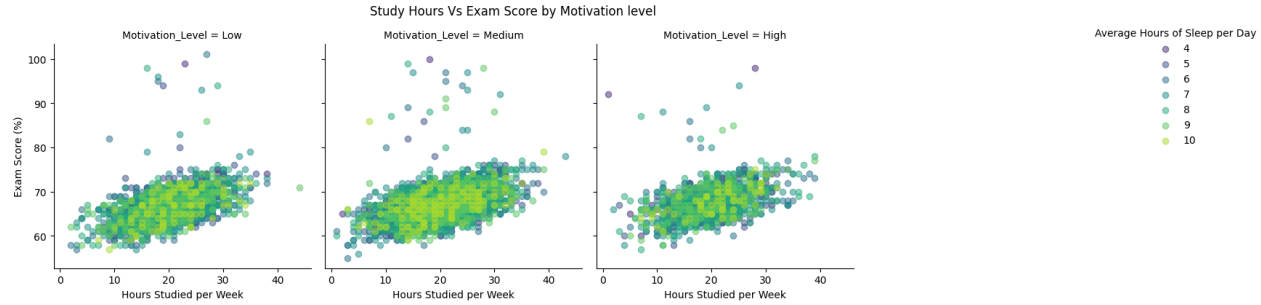



```
# Visualization 3: Histogram - Distribution of Exam Scores
'''
Code explanation:
This code uses matplotlib to create a histogram of exam scores to visualize the distri
scores are more narrow, and the distribution is more concentrated.
'''
plt.figure(figsize=(10, 6))
hist=plt.hist(df['Exam_Score'], bins=40, color='pink', edgecolor='black')
plt.title('Distribution of Exam Scores')
plt.xlabel('Exam Score (%)')
plt.ylabel('Frequency')
plt.grid(alpha=0.4)
plt.show()
```




```
# Visualization 4: Facet Plot- Study Hours Vs Exam Score by Motivation level
'''
Code explanantion:
This code utilizes seaborn to create a facet plot based on different motivation level
by the student. The hue of the plot is the average hours of sleep a student gets per
'''
plt.figure(figsize=(10, 6))
facet=sns.FacetGrid(data=df,col='Motivation_Level',col_wrap=3,height=4,hue='Avg_Hrs_S
facet.map(plt.scatter,'Hours_Studied_Per_Week','Exam_Score',alpha=0.5)
facet.set_axis_labels('Hours Studied per Week','Exam Score (%)')
facet.fig.suptitle('Study Hours Vs Exam Score by Motivation level', y=1.05)
facet.add_legend(title='Average Hours of Sleep per Day', bbox_to_anchor=(1.05, 1), lo
plt.show()
```

```
<Figure size 1000x600 with 0 Axes>
```



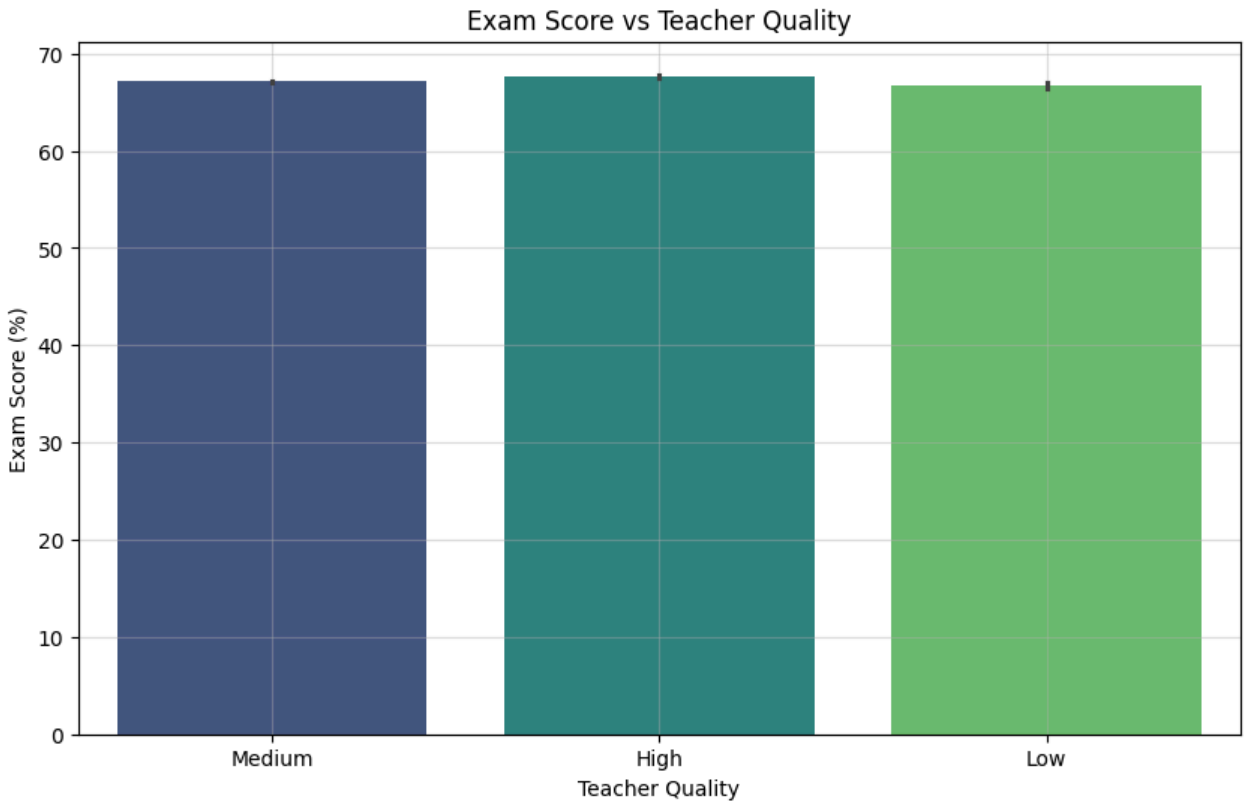

```
# Visualization 5: Bar graph- Exam Score vs Teacher Quality
'''
Code Explanation:
This code utilizes the bar graph from seaborn to visualize the relationship between a
by teacher quality and then the average exam score is calculated.
'''
plt.figure(figsize=(10, 6))
bar=sns.barplot(data=df, x='Teacher_Quality', y='Exam_Score', palette='viridis')
avg_exam_score_by_teacher_quality = df.groupby('Teacher_Quality')['Exam_Score'].mean(
bar.set_title('Exam Score vs Teacher Quality')
bar.set_xlabel('Teacher Quality')
bar.set_ylabel('Exam Score (%)')
```

```
bar.grid(alpha=0.4)
plt.show()
```

```
/tmp/ipython-input-364994217.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0

  bar=sns.barplot(data=df, x='Teacher_Quality', y='Exam_Score', palette='viridis')
```




```
# Visualization 6 - Heatmap (correlation matrix)
'''
This code uses the heatmap from seaborn to visualize the correlation matrix of the nu
are seperated frm the categorical attributes, and the correlation matrix is calculate
correlations.
'''
num_cols=['Hours_Studied_Per_Week','Attendance_Percentage','Avg_Hrs_Sleep_Per_day','P
plt.figure(figsize=(10, 6))
corr_matrix=df[num_cols].corr()
heat=sns.heatmap(corr_matrix, annot=True, cmap='crest', linewidths=0.5)
heat.set_title('Correlation Heatmap of Student Performance (Numerical Variables)')
plt.show()
```

```
# save notebook to pdf
```